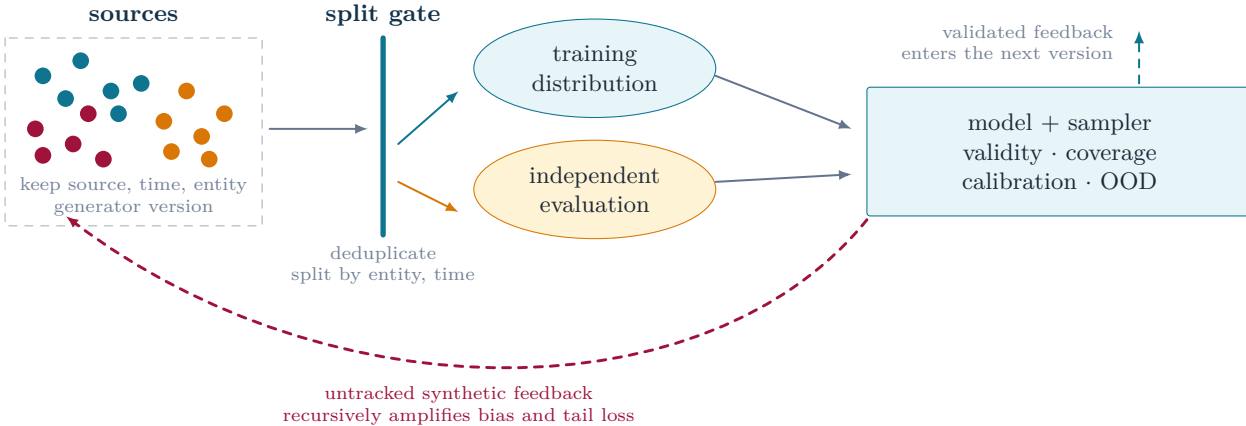A traceable data-to-deployment loop